



Building AI Agents with Multimodal Models

Part 1: Multimodal Data

Agenda

- Part 1: Multimodal Data
- Part 2: Intermediate Fusion and CLIP
- Part 3: Cross-modal Projection and OCR Pipelines
- Part 4: VSS and GraphRAG
- Part 5: Wrap-up & Assessment

Prerequisites

- Basic familiarity with convolutional neural networks (CNNs)
- Basic familiarity with a deep learning framework such as:
 - PyTorch
 - TensorFlow

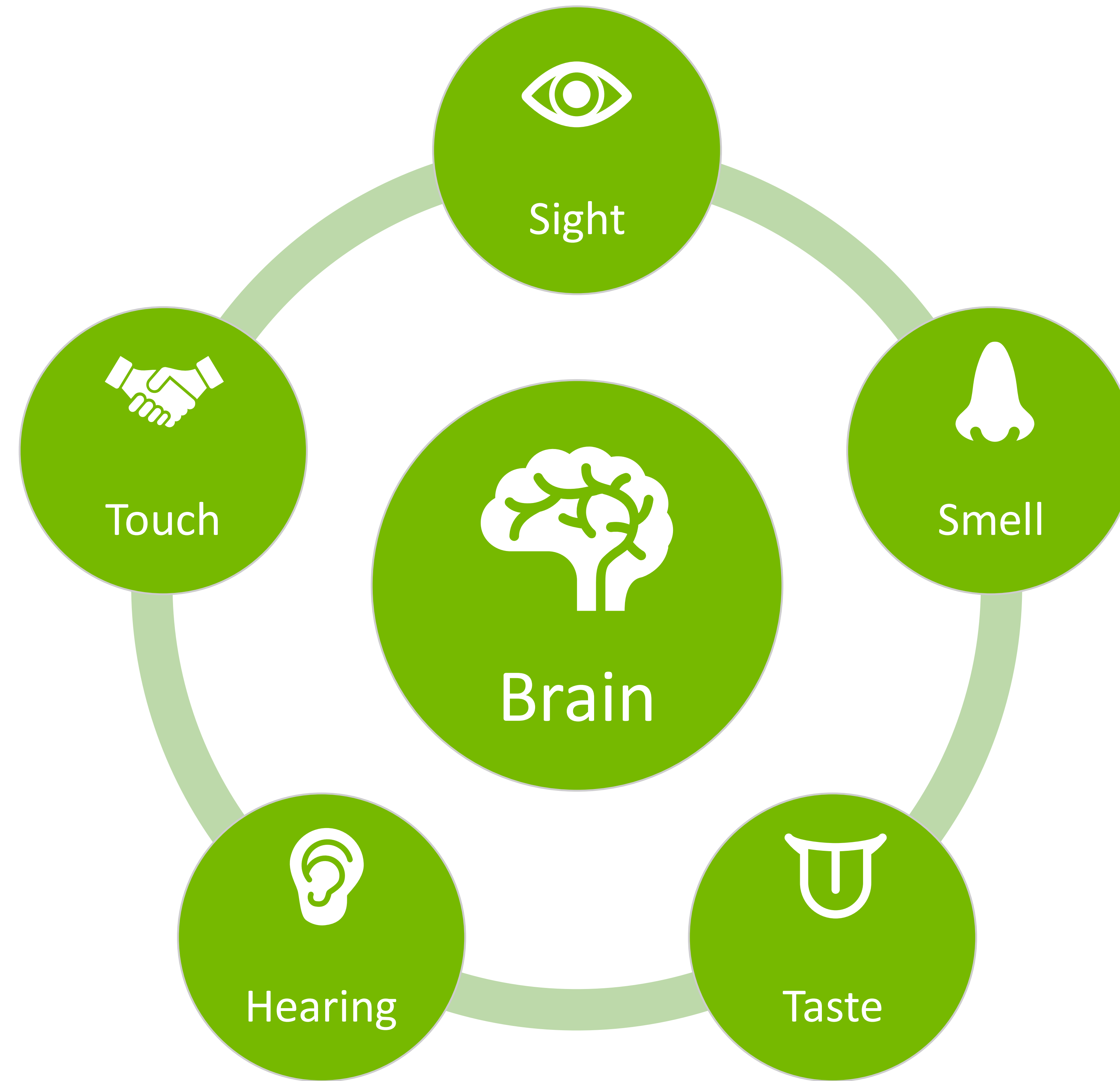




Multimodality

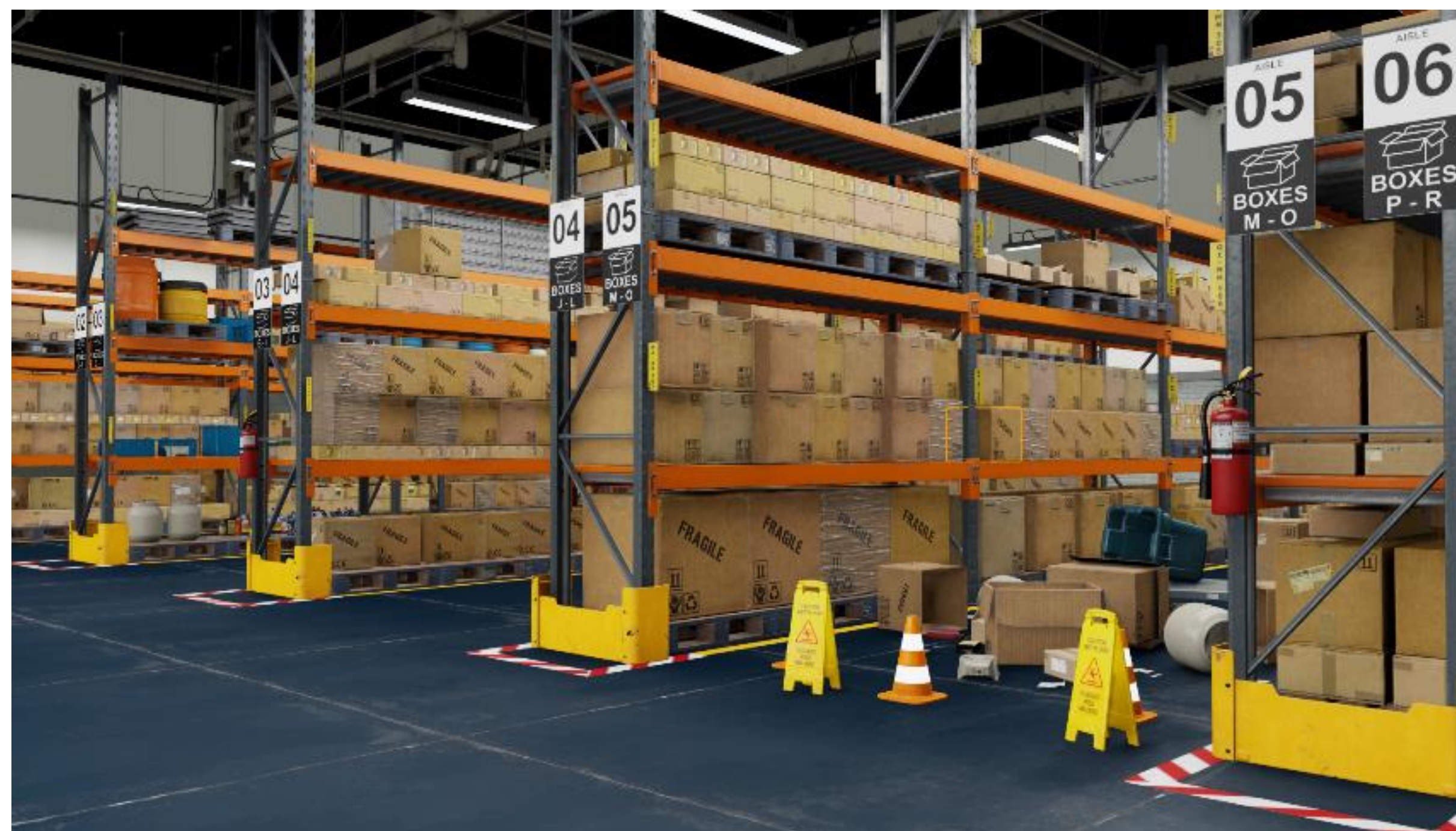
The Five Senses

How Humans Perceive

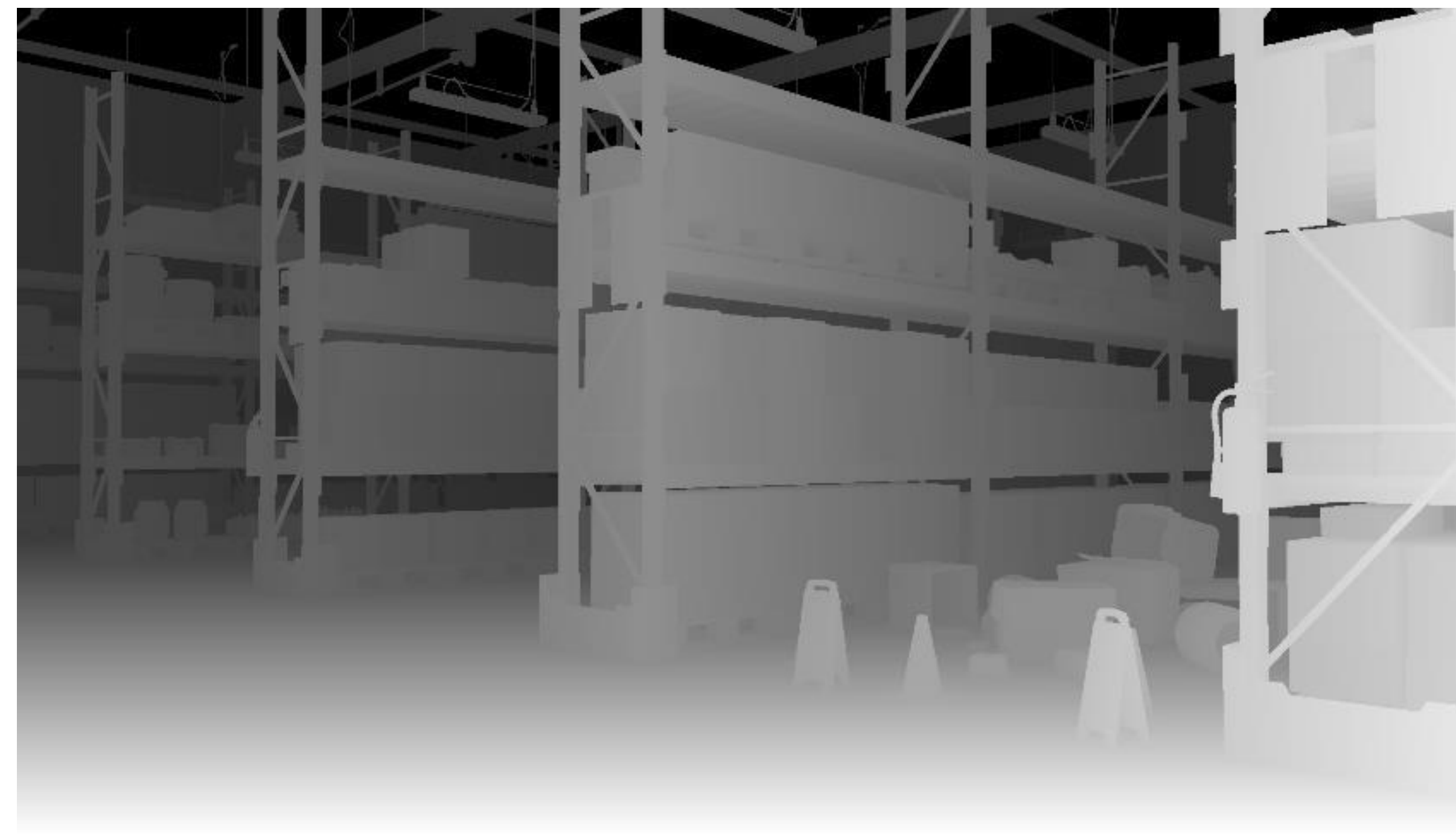


Sight

How A Computer Might See



RGB



Depth



Fisheye

Smell and Taste

How A Computer Might Smell and Taste

pH



Nutrition Facts

Nutrition Facts	
1 serving per container	
Serving size	140 g
Amount per serving	
Calories	100
% Daily Value *	
Total Fat 0.5g	1%
Saturated Fat 0g	0%
Trans Fat 0g	
Cholesterol 0mg	0%
Sodium 0mg	0%
Total Carbohydrate 24g	9%
Dietary Fiber 6g	22%
Total Sugars 15g	
Includes 0g Added Sugars	0%
Protein <1g	
Vitamin D 0mcg	0%
Calcium 0mg	0%
Iron 0.3mg	2%
Potassium 120mg	2%
* The % Daily Value (DV) tells you how much a nutrient in a serving of food contributes to a daily diet. 2,000 calories a day is used for general nutrition advice.	

Molecular Weight



Hearing

How A Computer Might Hear

Sheet Music

Spectrogram

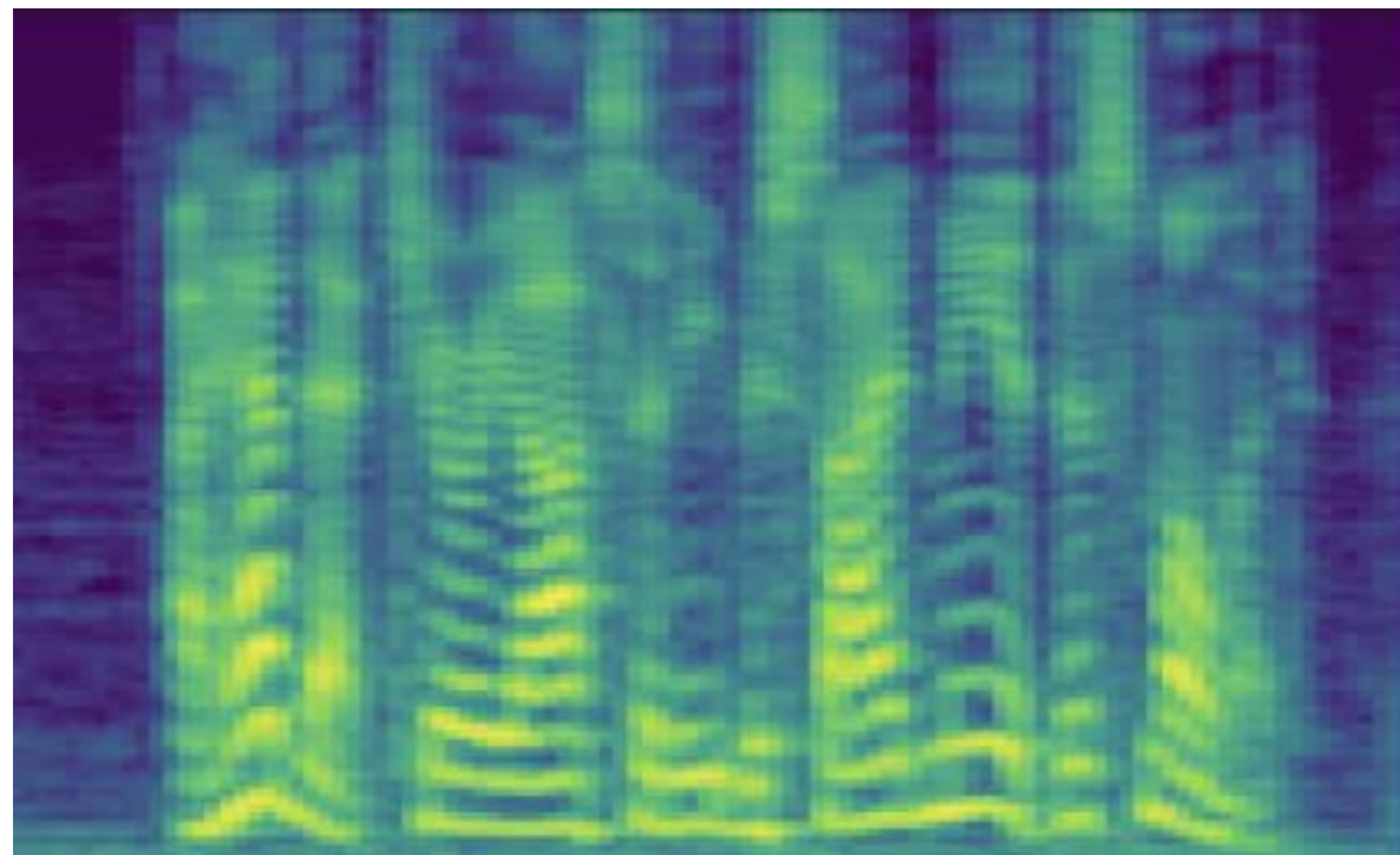
Dedicated to J. A. STERNAD.

Take Me Out To The Ball Game.

Words by JACK NORWORTH. Music by ALBERT VON TILZER.

Tempo di Valse.

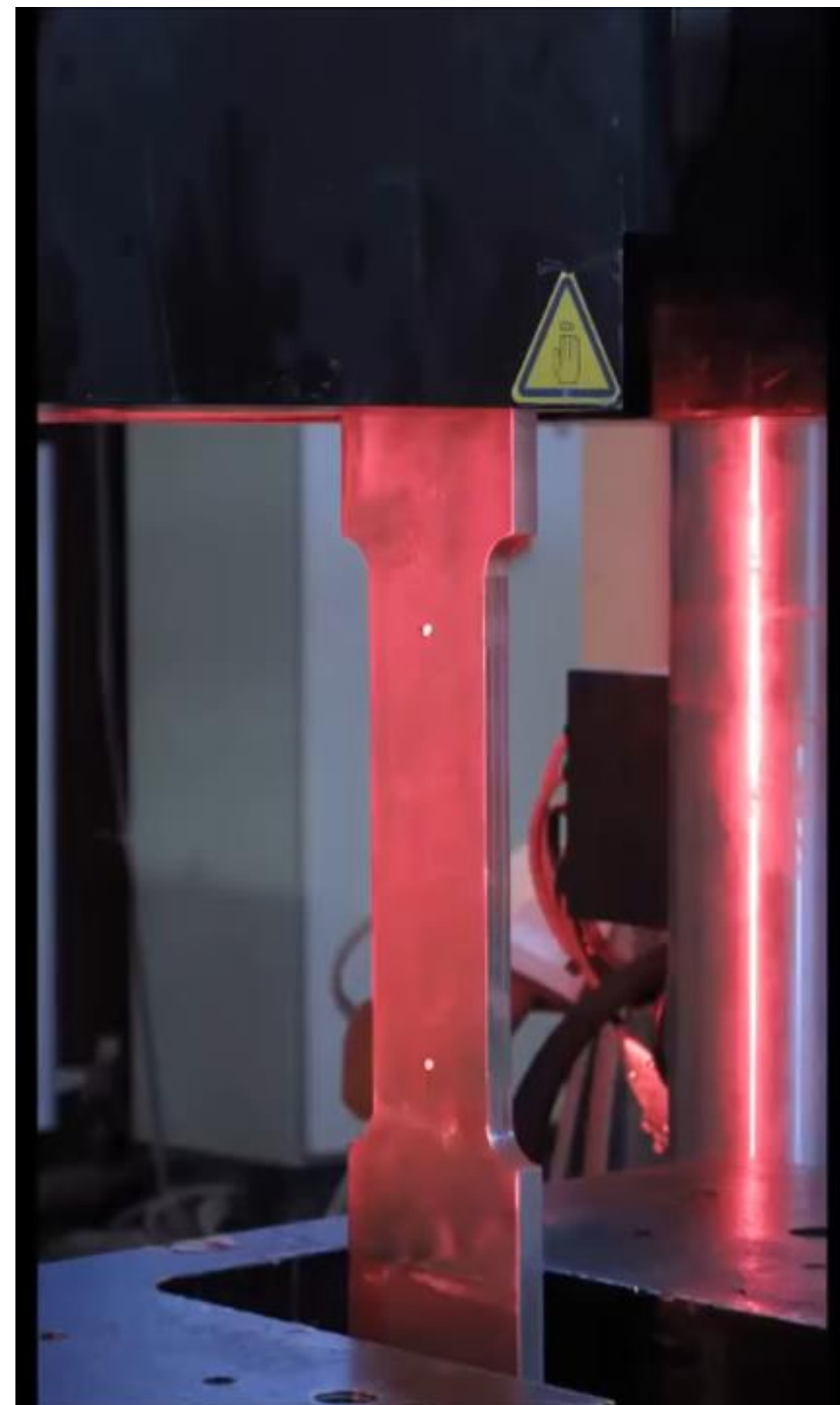
The image shows a page of sheet music for the song 'Take Me Out To The Ball Game'. It features a piano introduction in 2/4 time, followed by a vocal melody. The lyrics are: 'Ka - tie Ca - sey was / Ku - tie Ca - sey saw'. The music is in G major and 2/4 time.



Touch

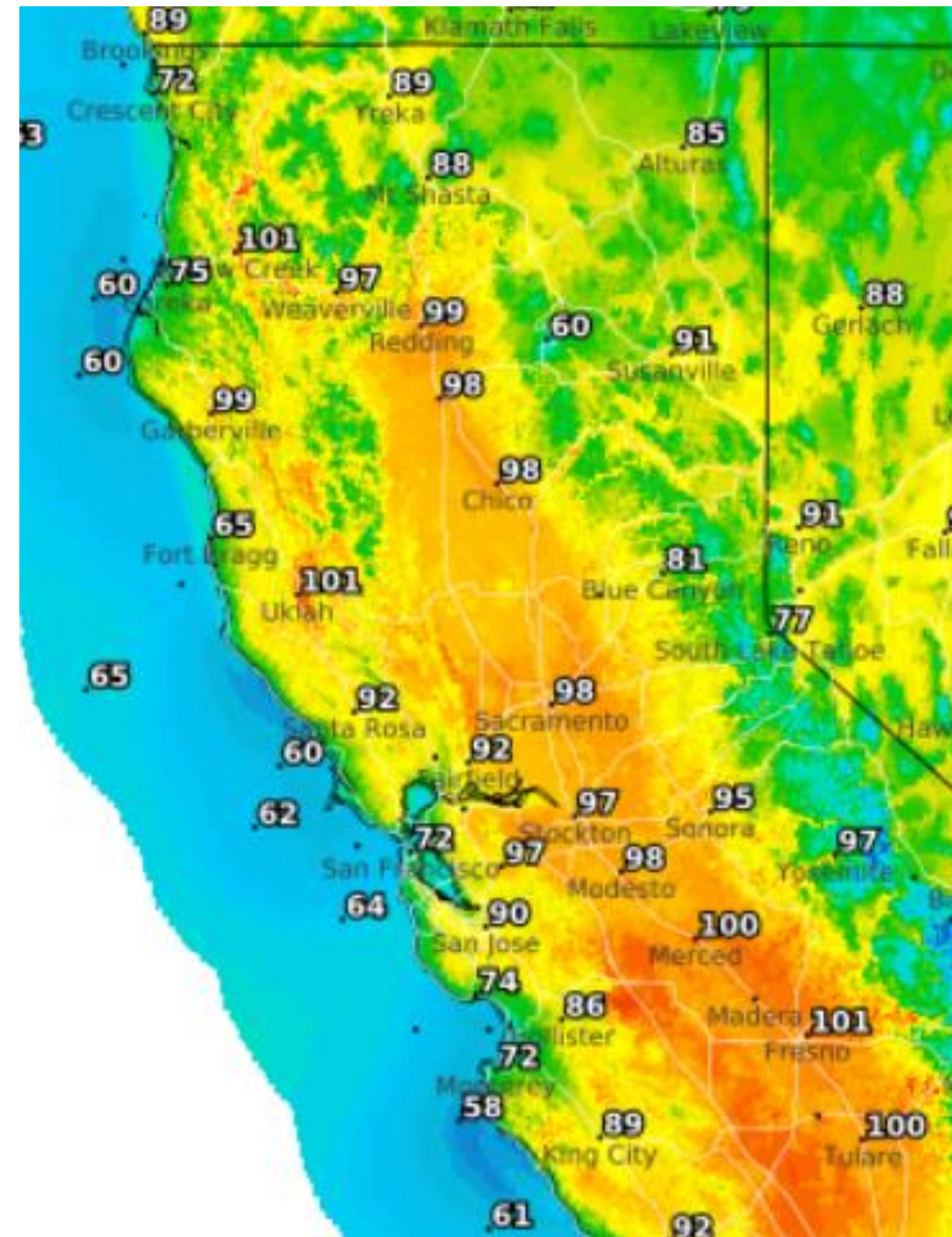
How A Computer Might Feel

Tensile Strength



By [Christian V.](#)

Weather Maps



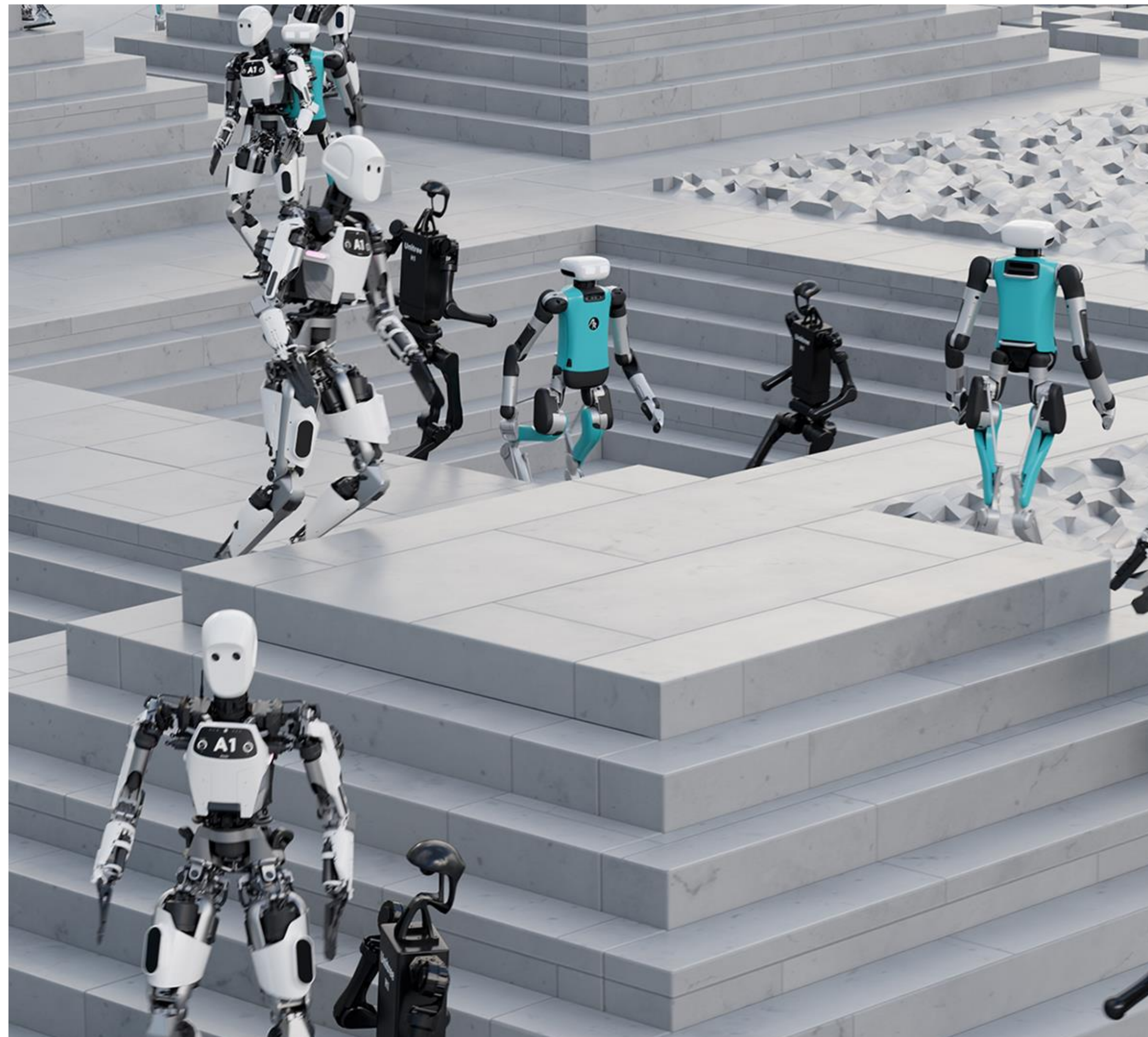
Weight



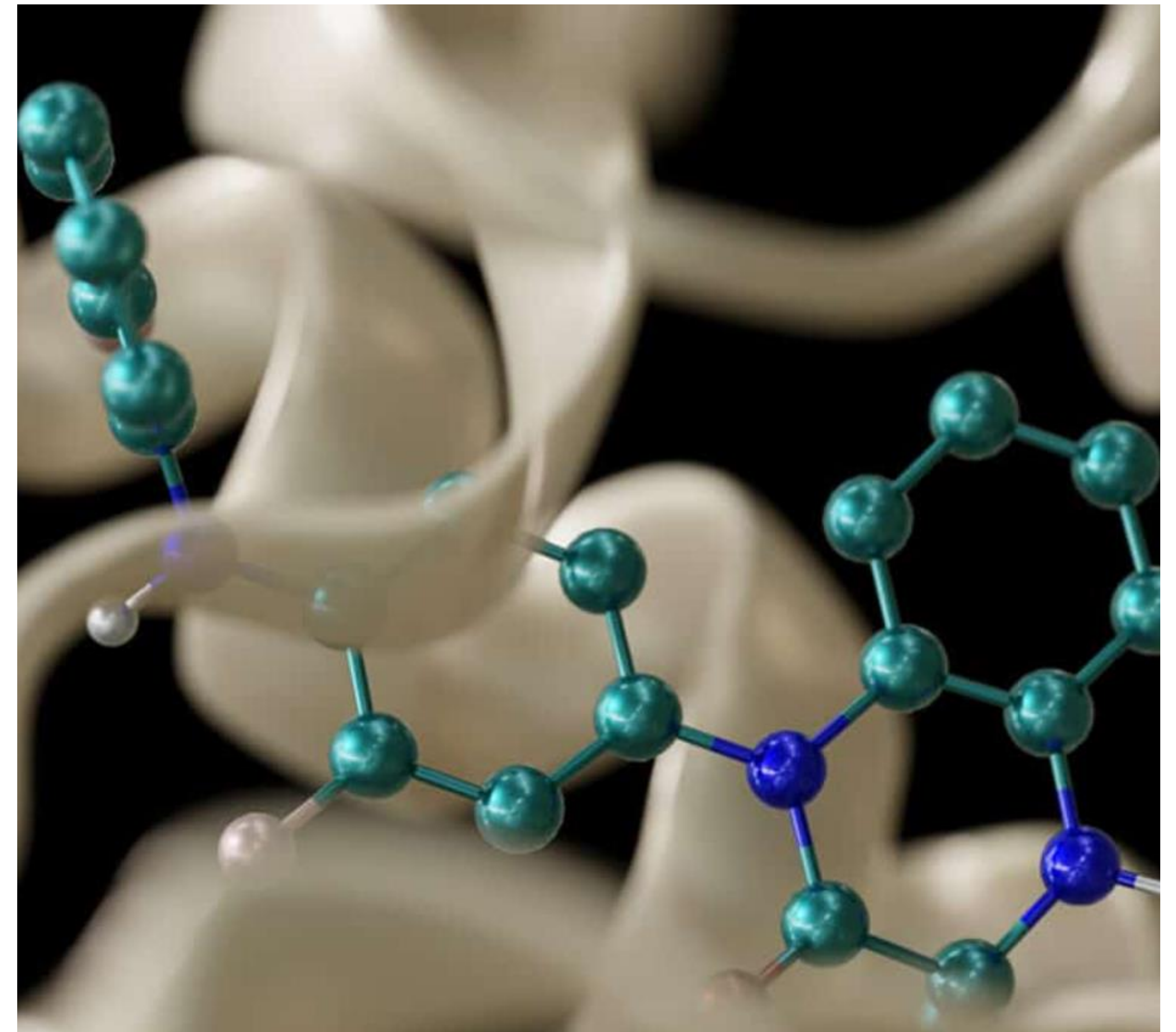
Multimodal Applications

Common Multimodal Use Cases

Robotics



Biology



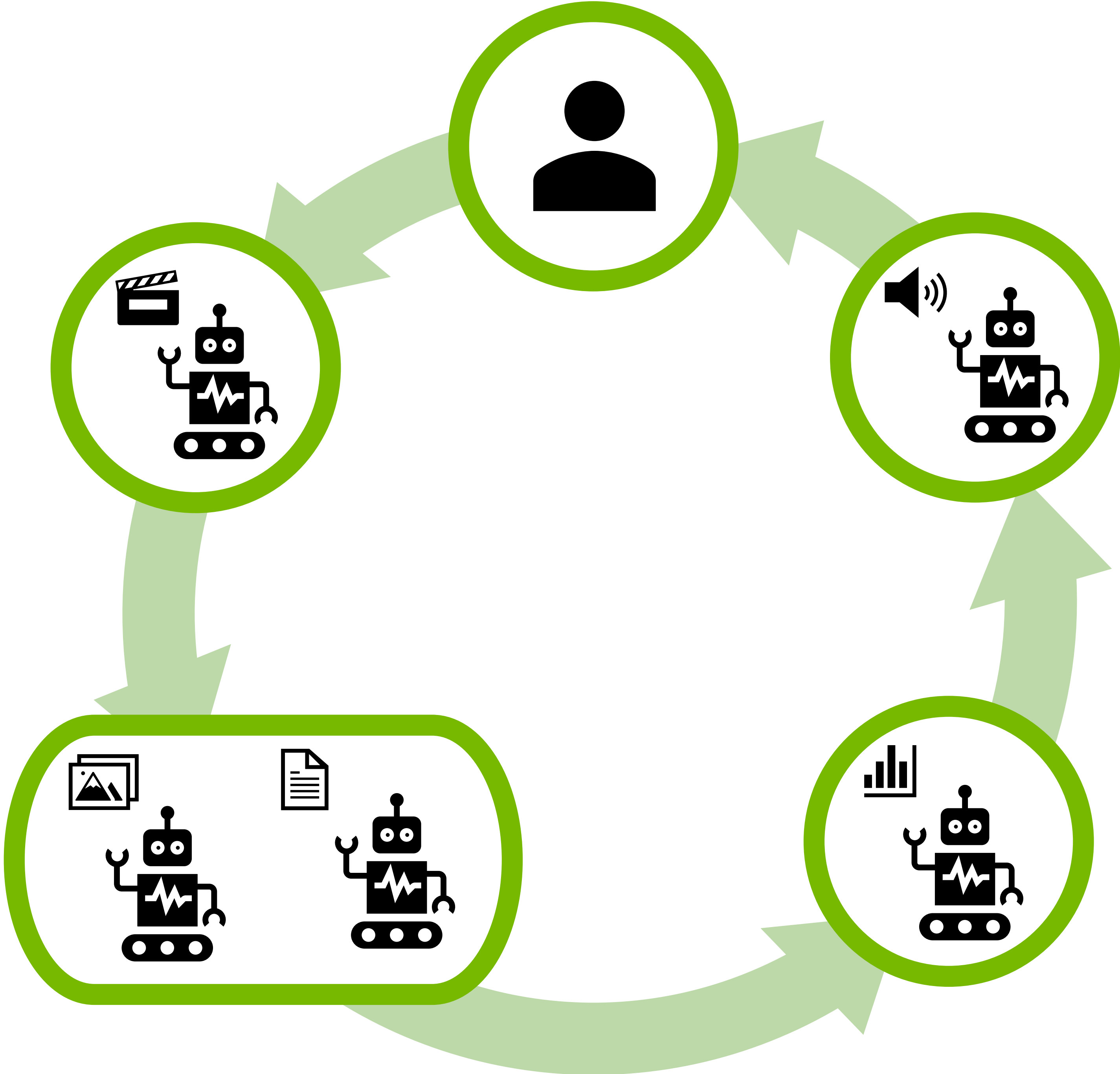
Multimodal vs Orchestration

Common Multimodal Use Cases

Modality



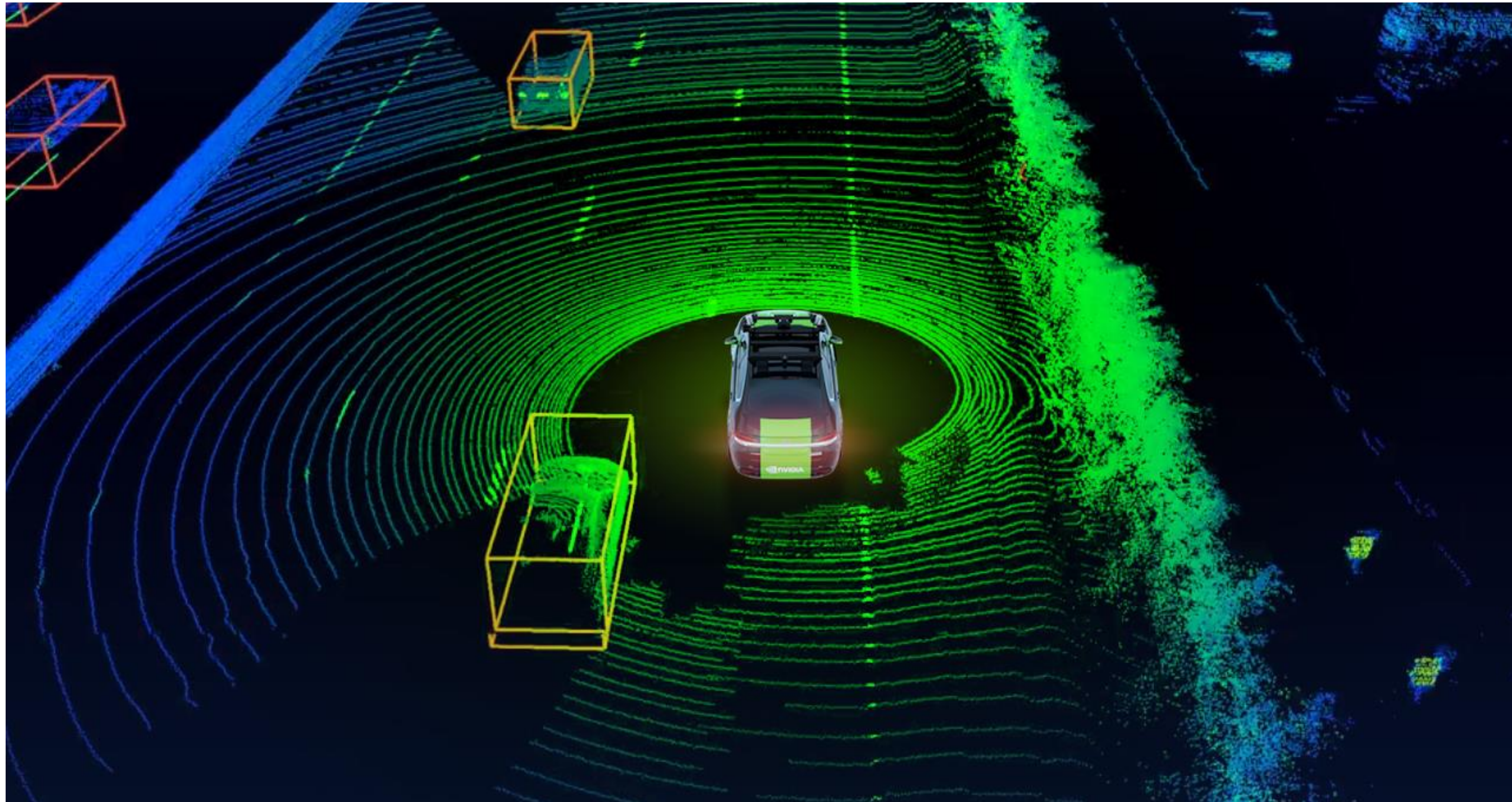
Orchestration





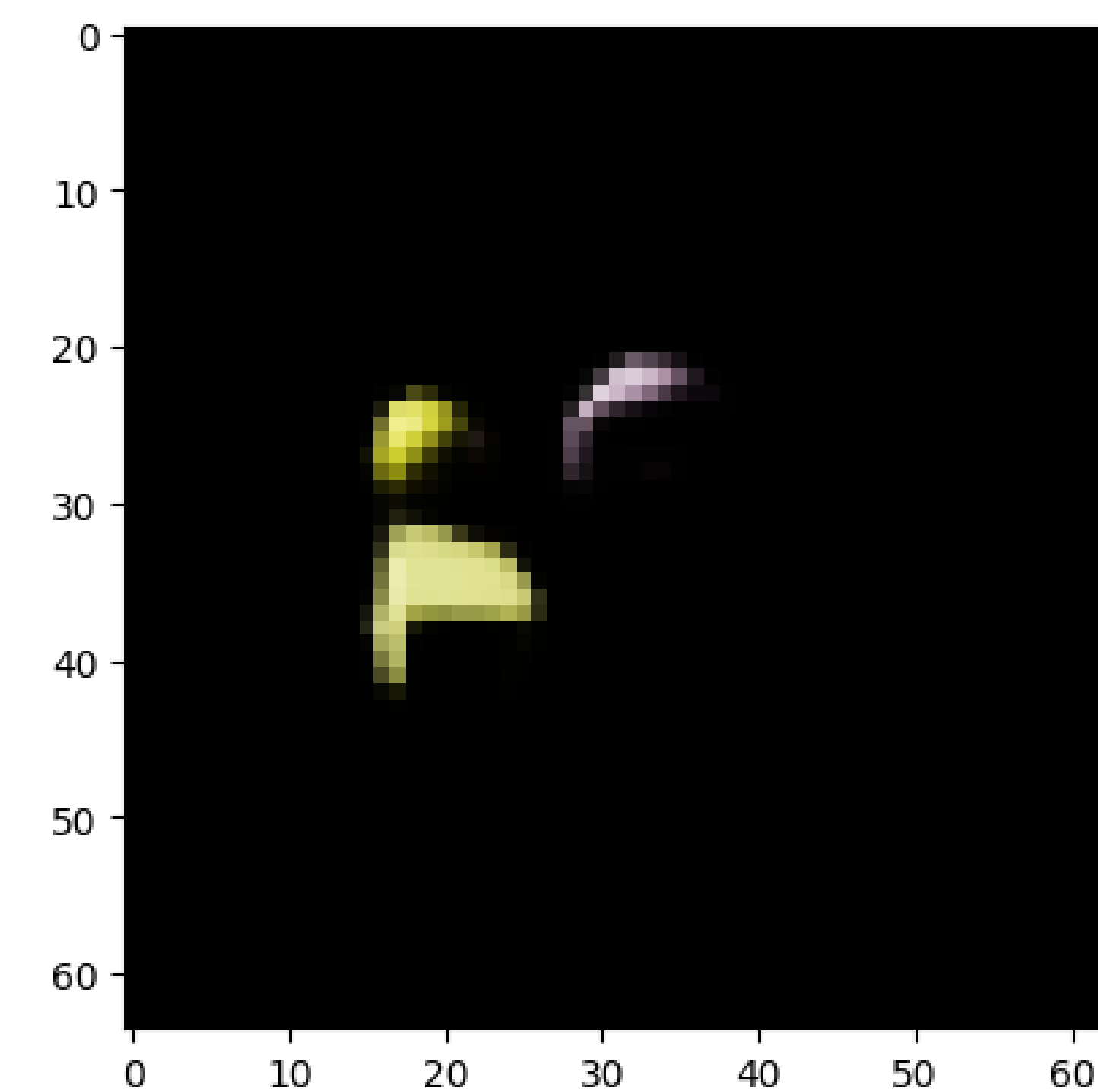
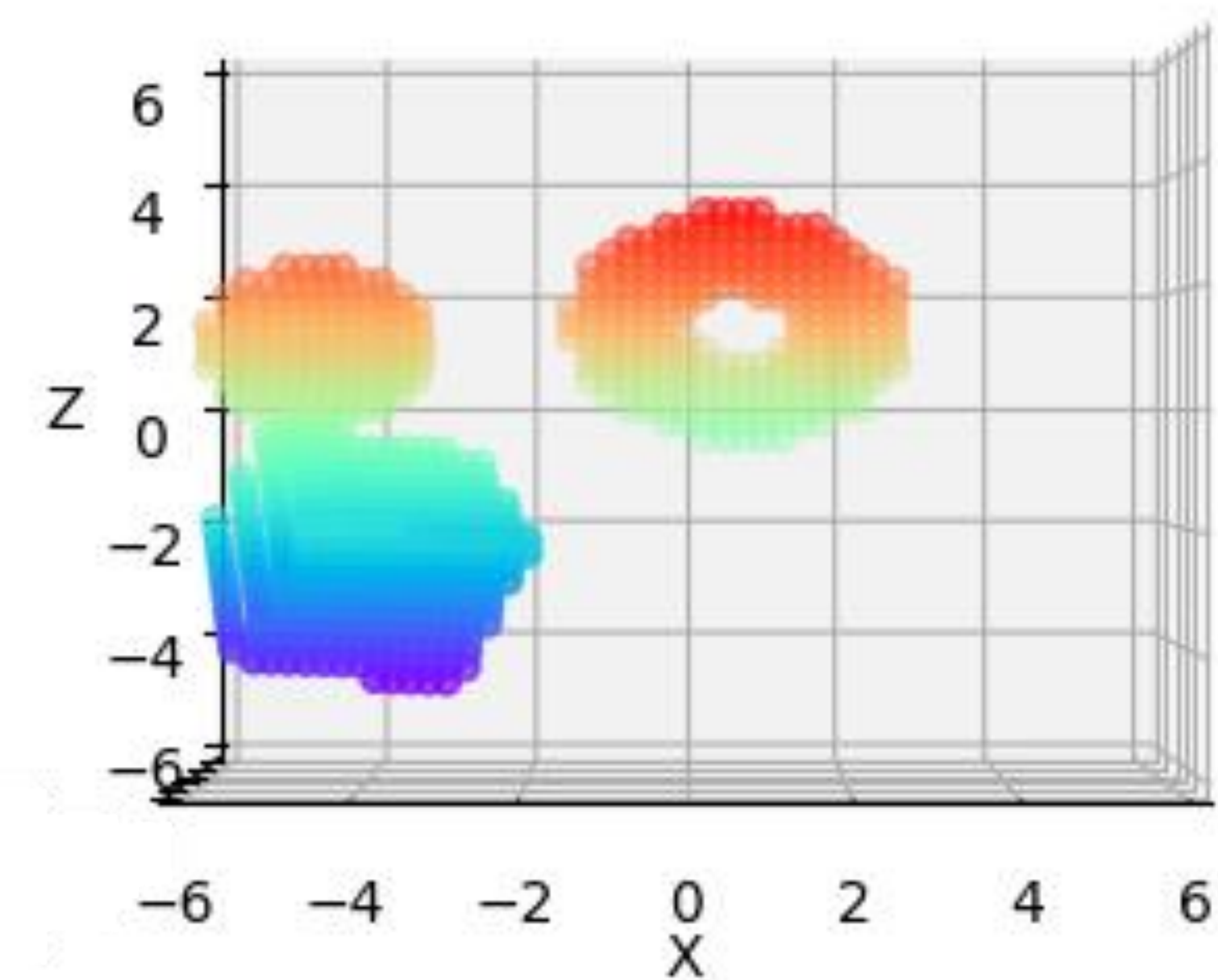
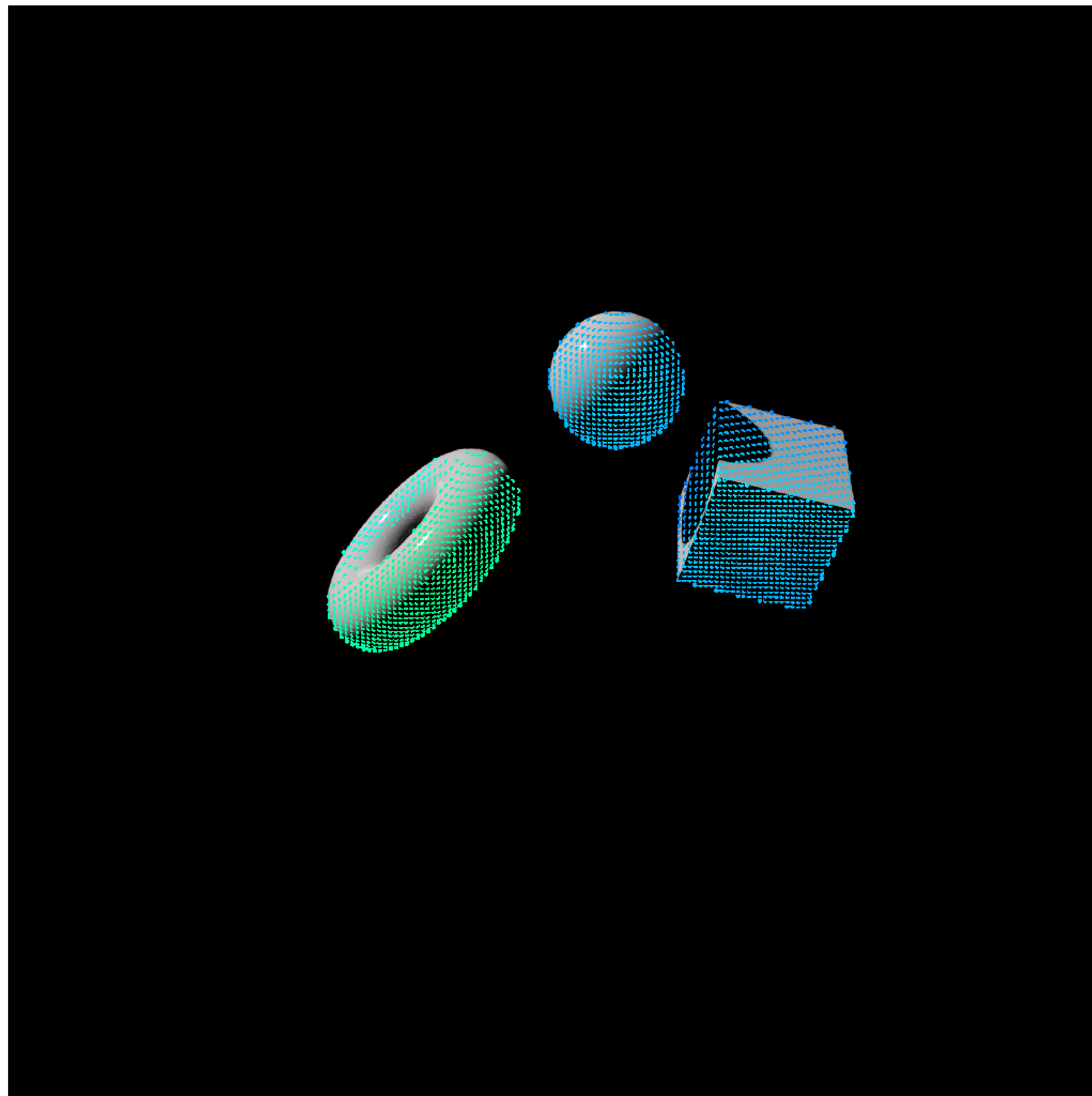
Lab 1a – Early and Late Fusion

Using LiDAR Data



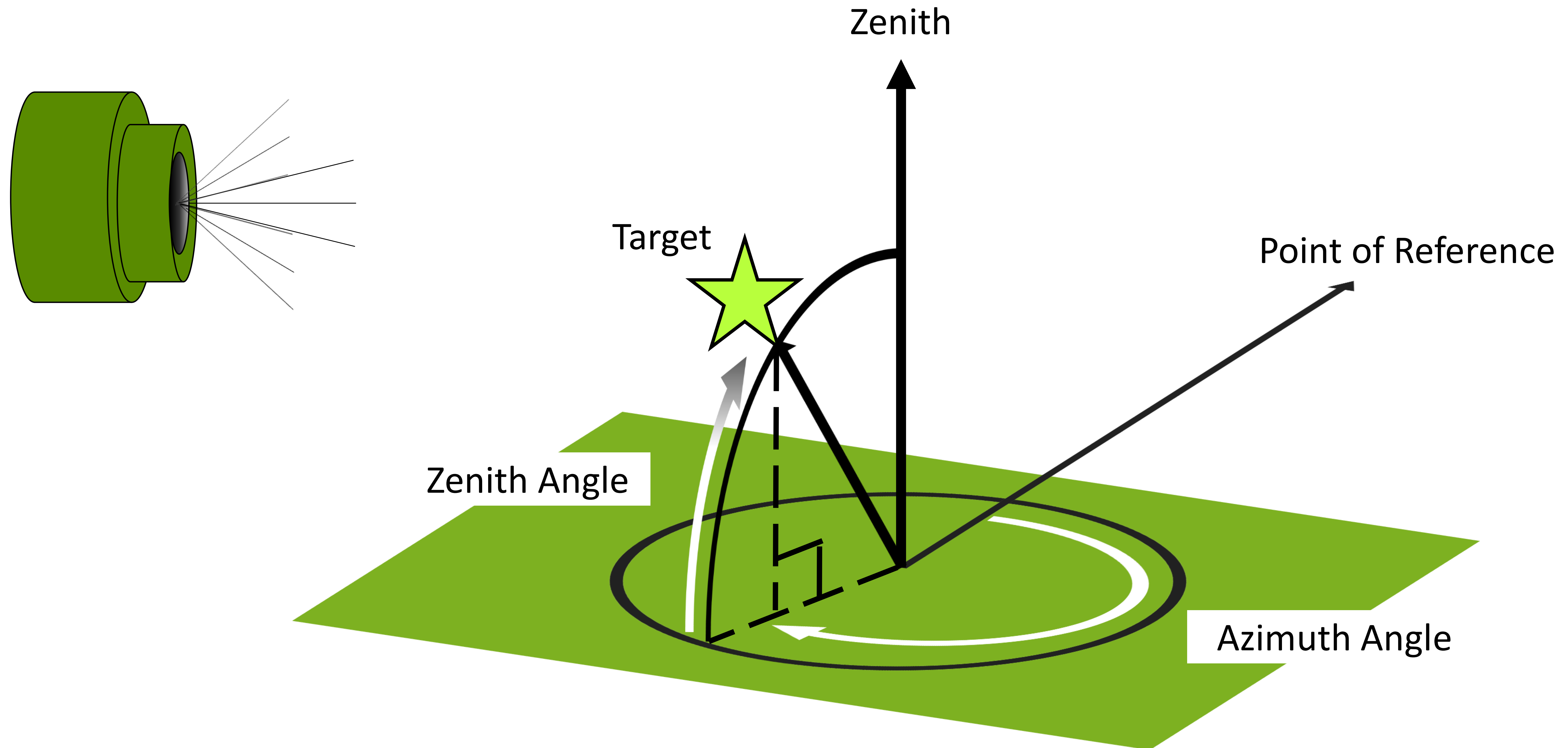
LiDAR vs Camera

Common Multimodal Use Cases

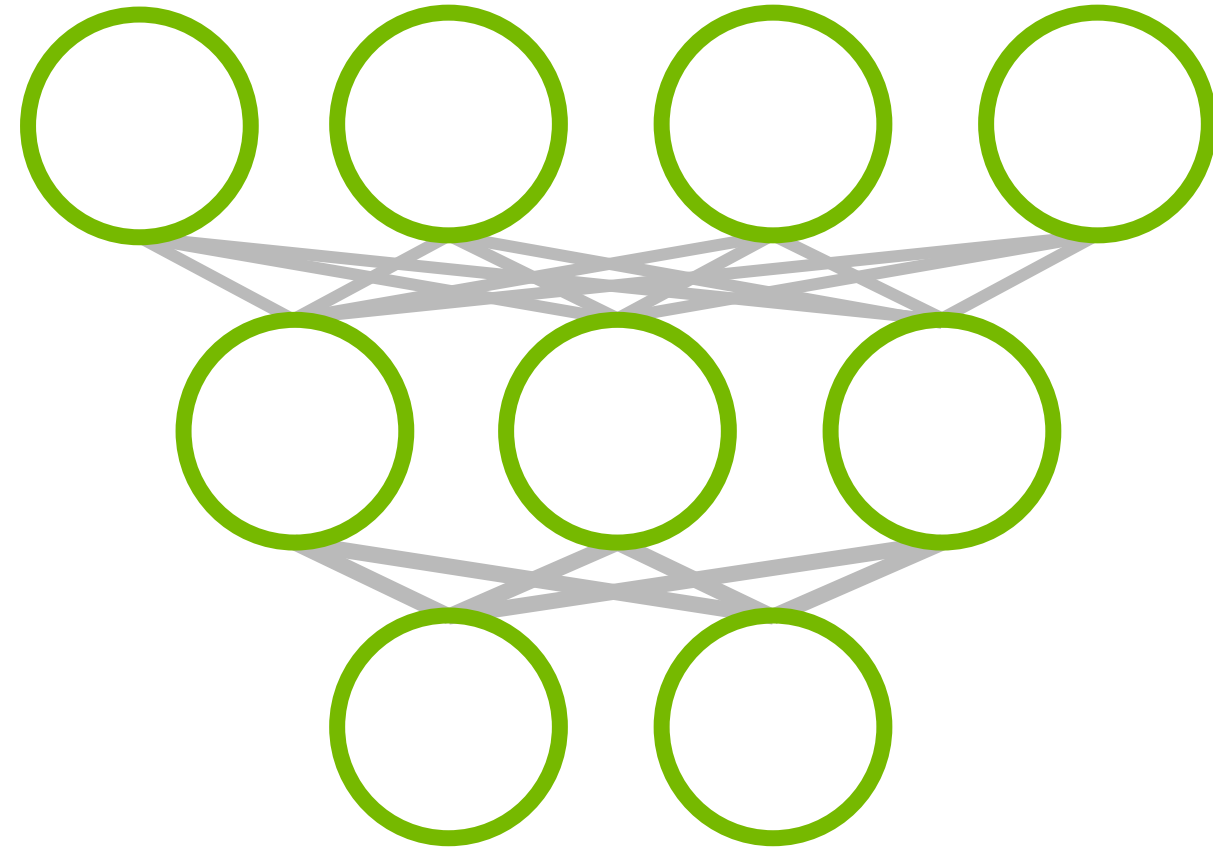
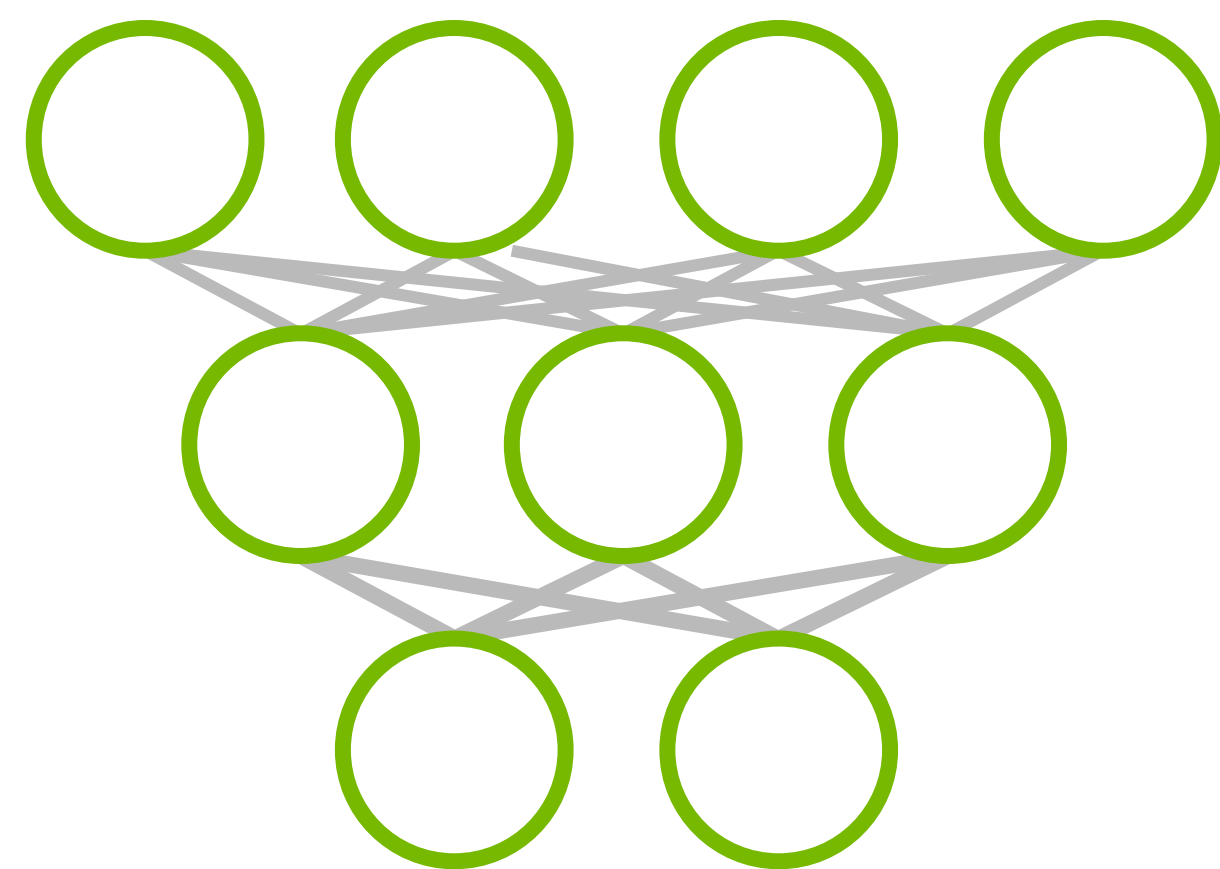
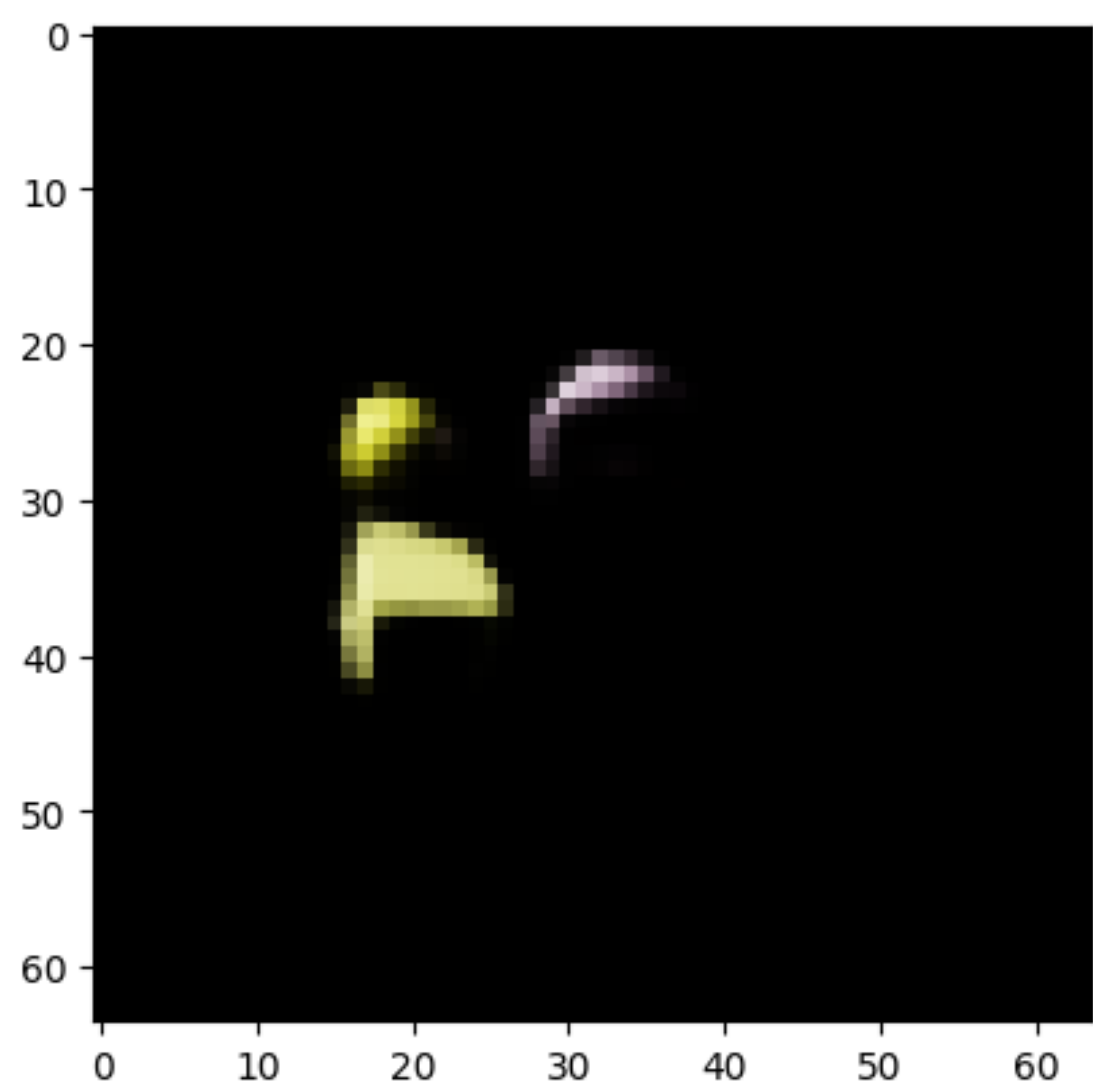
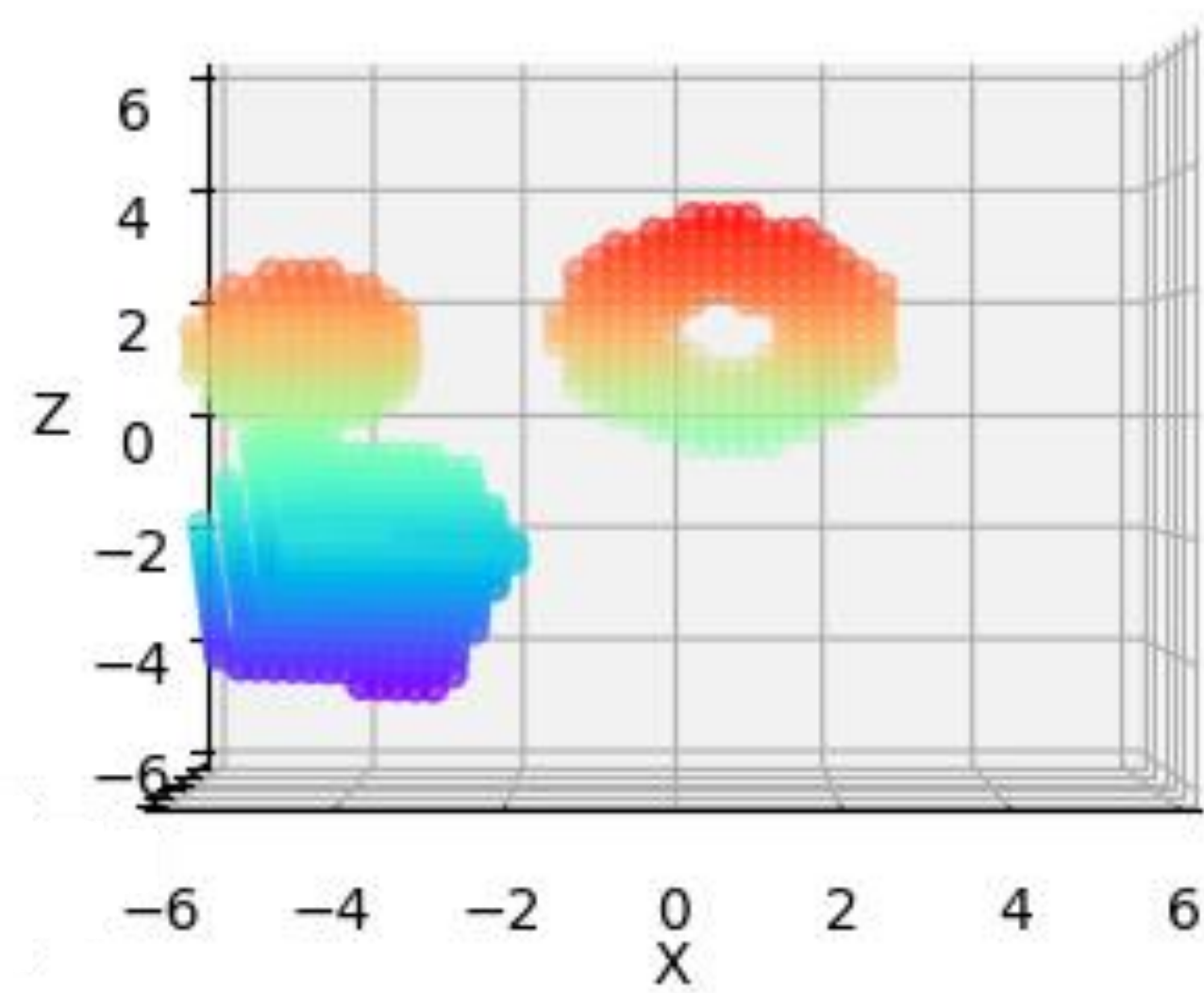


Azimuth and Zenith

Finding Stars

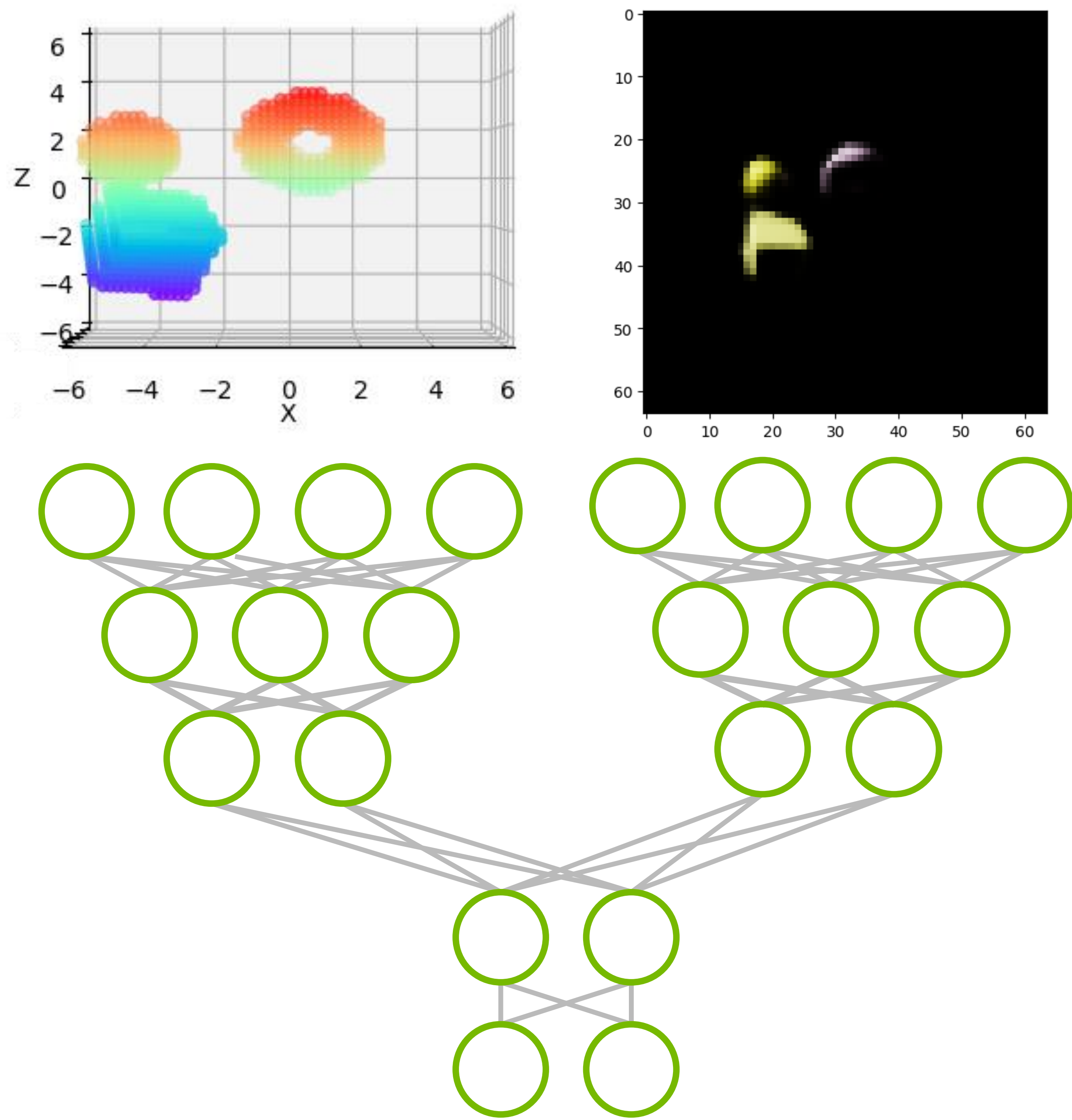


Fusion Techniques



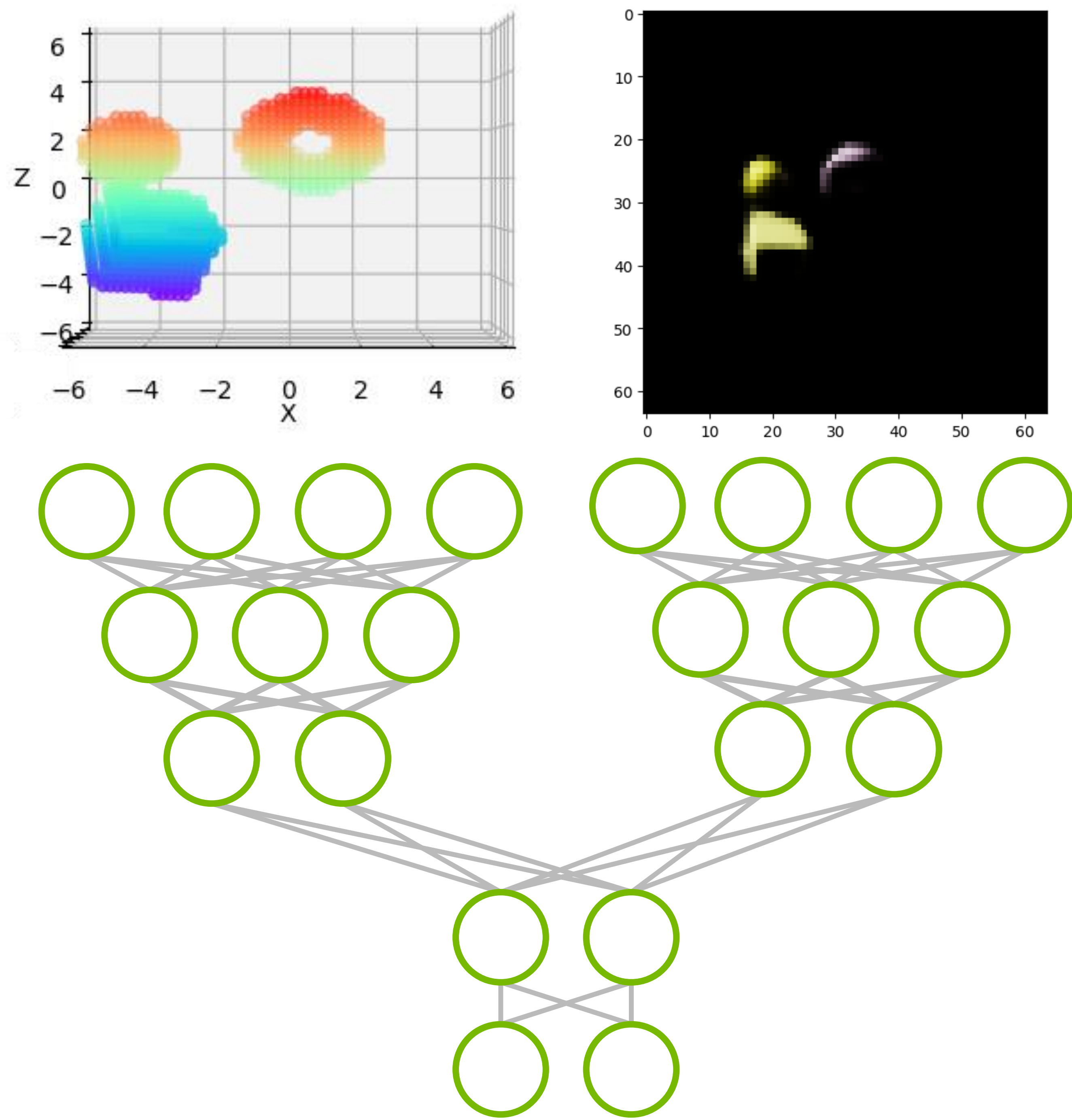
Fusion Techniques

Late

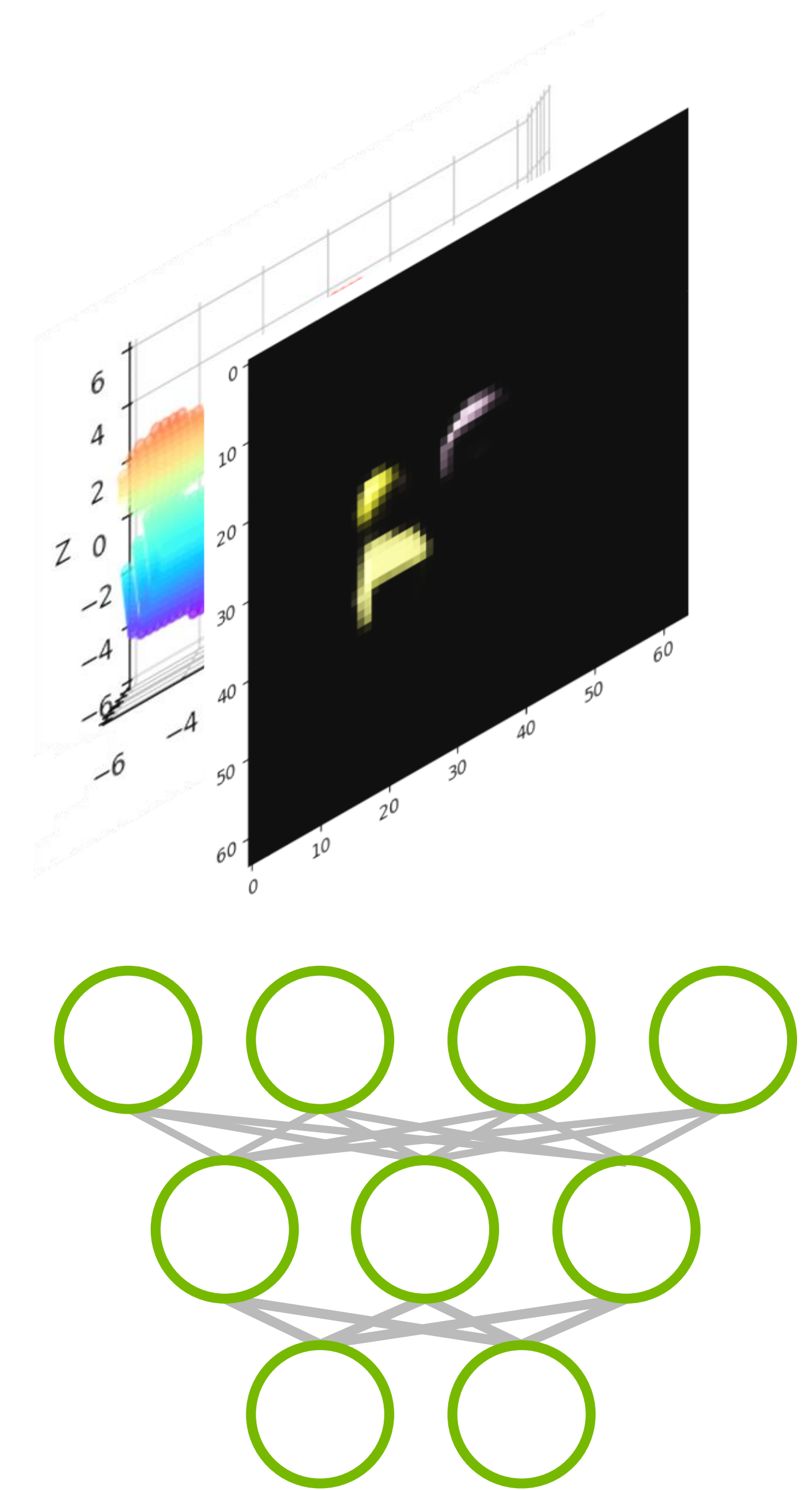


Fusion Techniques

Late



Early





Lab 1b – Exploring Modalities

“

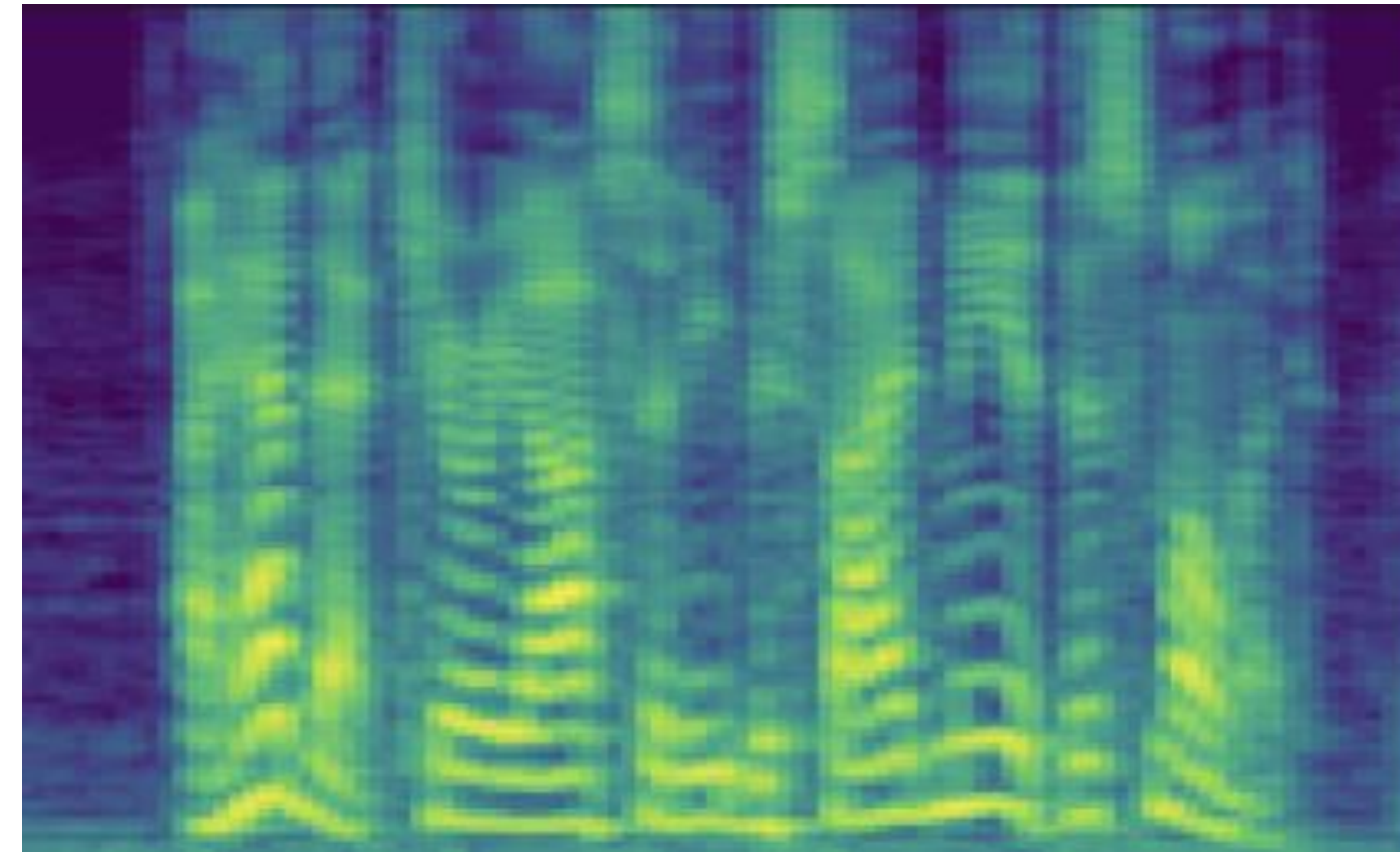
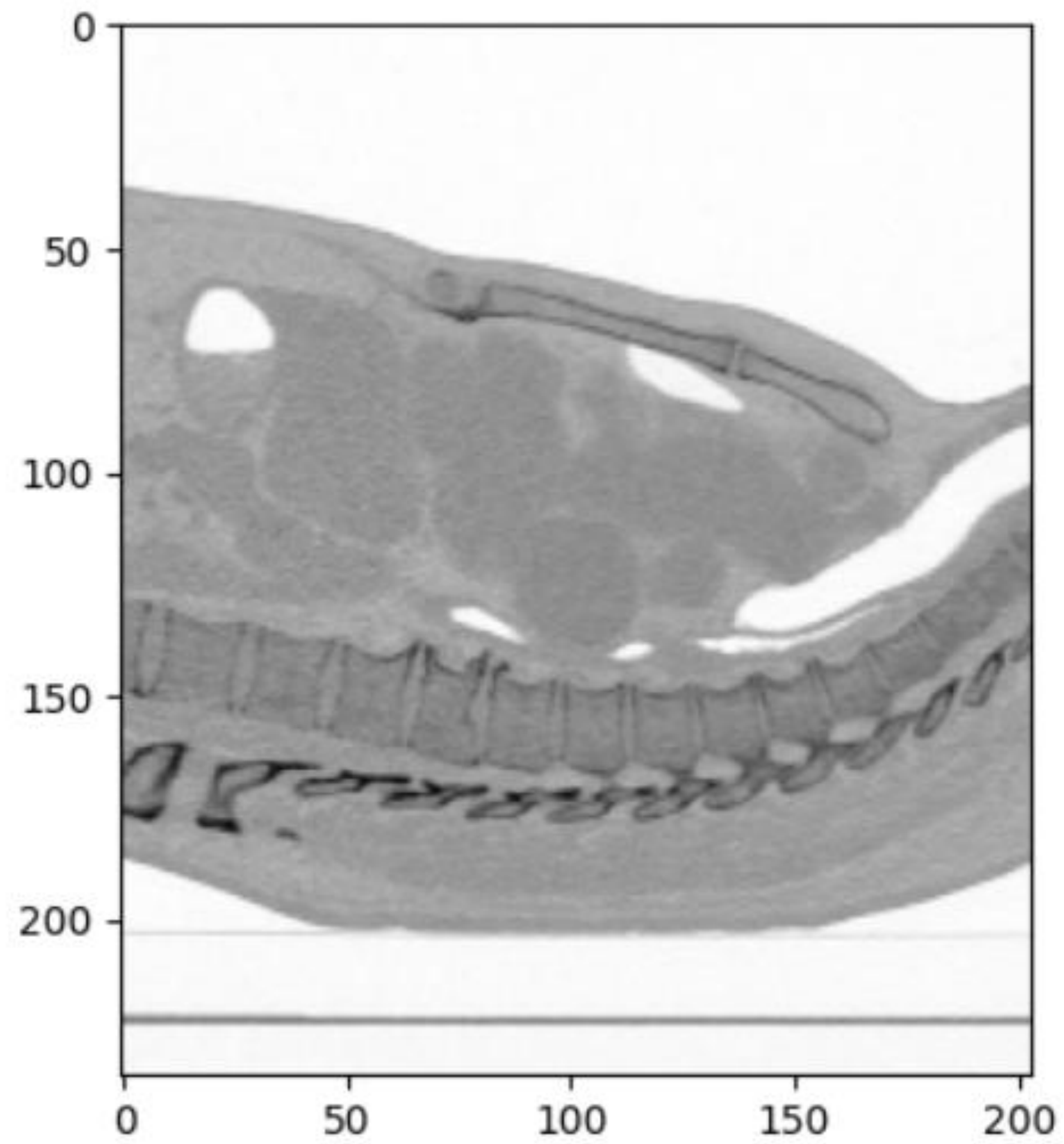
Hypothesis:

Any data that can be turned into a matrix can be used for neural network reasoning

”

Atypical Data

CT Scan



Spectrogram



Magnetic Data



Let's Get Started!

